

MOHAMMED ZAKWAN M

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TECHNICAL SKILLS

Core Programming: Python (Proficient), C, Java, SQL, JavaScript, HTML/CSS (Fundamentals)

Computer Vision & AI/ML: Object Detection (YOLO Architectures v1–v12, YOLO26), From-Scratch Model Implementation, Image Processing, Deep Learning Pipelines, Neural Networks

Frameworks & Libraries: PyTorch, OpenCV, NumPy, Pandas, Matplotlib

Developer Tools: Git, GitHub, VS Code, Google Colab, Jupyter Notebook, Antigravity

Core CS Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems

EDUCATION

Bachelor of Technology in Computer Engineering (Artificial Intelligence & Machine Learning) 2023 – 2027

Presidency University, Bangalore, Karnataka | CGPA: 7.24 / 10.0

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Software Engineering, Storage Architecture

PROJECTS

Aerial & UAV Object Detection & Tracking System

Python, PyTorch, YOLO, OpenCV, TensorRT, ONNX | github.com/zakwanmunawar/AeroDrop-Tracker

- Trained a YOLO26n detection model (2.37M parameters) on a 12-class aerial dataset with 86,000+ annotations, reaching 22.1% mAP@50.
- Optimized the inference pipeline on an NVIDIA Tesla T4 GPU for sub-3ms latency, sustaining over 330 FPS.
- Built the preprocessing pipeline from scratch, covering frame resizing, bounding-box normalization, and data augmentation.
- Integrated OpenCV for live video inference, enabling frame-to-frame object tracking with minimal lag.

Construction Worker Safety & PPE Compliance Detector

Python, YOLO, OpenCV | github.com/zakwanmunawar/worker-safety-system

- Fine-tuned a 3M-parameter YOLO model to detect site safety violations, including falls and missing helmets or jackets, reaching 97.0% mAP@50 and 96.5% precision.
- Deployed the pipeline on a Tesla T4 GPU at sub-9ms latency, supporting real-time compliance alerts.
- Added color-coded bounding-box overlays in OpenCV to visually flag missing safety gear in the live feed.

CERTIFICATIONS & ACHIEVEMENTS

YOLO Architecture Implementation & Research Series (v1–v12, YOLO26)

PyTorch & Google Colab

- Rebuilt YOLOv1 from scratch in PyTorch, then worked forward through later versions to understand what changed at each step.

Virtual Internship Certification in Python Programming – CodSoft

2025

- Completed a 4-week virtual internship, writing modular Python scripts and practicing core OOP and algorithmic concepts.

Data Structures & Algorithms in Python – NeetCode 75

- Worked through the NeetCode 75 problem set in Python, covering core patterns including two pointers, sliding window, graphs, and dynamic programming.

Top 70 Selection (out of 500+ Teams) – University Innovation & Project Review (IPR)

- Built the hardware-software integration for an autonomous robotic car arm, one of 70 teams selected out of 500+.